DO NOW — RECALL

Interrupts and service routines

Name: _____ Class: _____ Date: _____

1. Define fetch-decode-execute.

2. What causes a process to enter Waiting?

3. What does an OS scheduler choose?

4. Simplify (A∧B)∨(A∧¬B), showing factoring.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Read, interpret, execute an instruction.

2. Model answer

Unavailable I/O or required event.

3. Model answer

Which Ready process receives CPU time.

4. Model answer

A∧(B∨¬B)=A∧1=A.

Vocabulary — know these terms

interrupt — CPU attention request

ISR — Interrupt Service Routine

context — State for safe resume

polling — Repeatedly checking event state

NEW LEARNING

Read, annotate and explain

Understand

An interrupt signals an event such as device completion to the processor.

Trace

The CPU saves sufficient execution context before servicing the event.

Explain

An ISR handles the interrupt, acknowledges it, then restores normal execution state.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Keyboard triggers an interrupt during a text editor. Trace the stages.

Predict first

The ISR processes keyboard input before restoring the prior program state.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define interrupt, ISR and one source of interrupts. [4]

Question 2

Why does CPU context need to be saved/restored around a device ISR? [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare polling and interrupt-driven I/O for infrequent device input. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An interrupt handler can overwrite original registers without preserving any state.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.